# THICKSENSE

Vibration-Sensing Smart Spoon for Drink-Consistency Assessment

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## PROBLEM STATEMENT

### Growing Elderly Population

**2 in 5** elderly home residents experience *dysphagia* (difficulty swallowing)

### Inconsistent Thickness

Water, Milo and soup can require different thickener preparation

### Risk of Incorrect Consistency

Too thin: aspiration risk
Too thick: reduced fluid intake




## VALUE PROPOSITION

### Confidence in Every Cup

Helps caregivers check drink consistency before serving

### Simple and Intuitive

Dip. Press. Check. Three simple steps, designed for everyday care

### Market Opportunity

Singapore aged 65+:
**13.1%** → **20.7%**
*(2015)* *(2025)*

## SENSING PRINCIPLE



### Vibration Input

Motor vibrates the spoon

### Piezo Sensor

Detects changes in vibration

### Microcontroller

Processes the sensor signal

### Consistency Output

Displays an estimated result

## PROTOTYPE DESIGN

**A coin motor excites the spoon; a piezo sensor detects its response after damping.**




*Proof of Concept*

*Integrated PCB and Sensors*

*Smart Spoon Concept*

## CAREGIVER WORKFLOW

1. **DIP** Place spoon in drink
2. **PRESS** Start measurement
3. **CHECK** Read consistency result

## VALIDATION & TESTING

### Benchmark consistencies

Compare readings with IDDSI flow tests.

### Check Reapeatability

Repeat measurements to assess consistency..

### Stakeholder Words

"We know how much to add for water. But how about for milo or soups?" - HCA Nurse

## PROTOTYPE TO PILOT

**Prototype hardware → Validation → Caregiver Pilot**

Estimate consistency → coin motor + piezo hardware → IDDSI, repeatability and usability

**Prototype testing** → **Signal analysis** → **Design refined** → **Caregiver pilot**

## COMMERCIALISATION & IMPACT

### Prototype Cost vs Target Price

Less than S$100 parts → S$15 selling target

### Designed for Scale

Integrated PCB → fewer components → volume production

THICKSENSE | Safer Drinks. Brighter Days. | ~3 s |